

GDC is based on top of the GNU Compiler Collection, LDC is based on LLVM and DMD is the reference compiler created by the same person that made D, Walter Bright. GDC and LDC are great compilers since they produce more performant binaries than DMD does but they are difficult to set up in Windows than DMD. We're focusing on the Windows OS in this article.

Go to the DMD download page (<http://dgit.in/DMDDownload>) and get the installer for Windows. Once you've downloaded the zip file, simply unzip it in any location and add the path for the binaries in the environment variables for Windows. It should be "dmd2\windows\bin". This is only if you are using just one version of DMD, if there are more than one installations of DMD then there will be a conflict during calls and you'll run into a few errors or warning over version differences. You can always use the installer and get it done with since it too does exactly the same thing. Once you have the path configured, open up command prompt and type 'dmd' and press enter. Invoking DMD in this manner without any extra arguments will cause it to display the version number along with all the valid command line options. If you do see those then your installation went fine and you can begin programming in D.

You can also use IDE bundles like DDT, Mono-D, Visual-D and Code::Blocks. You can now start programming in D. All source files are to be saved with a .d extension. As is customary with any new programming tool, here is a hello world program for D.

INLINE BOX CODE:

```
import std.stdio;

void main() {
    writeln("Hello World!");
}
```

Once you have saved the program (let's call it hello.d), simply navigate to the folder in a command prompt window and type "dmd hello.d". Your console should have a new line as Hello World!, if this is not the case then there was a problem in locating the compiler

binary, recheck the installation and ensure that the binaries have been added to the path.

Speed

D is as fast as C++ if you use the right compiler and also the correct arguments when compiling on a platform. Using the default arguments can lead to code that is ridiculously slow. Here is a little benchmark that was run on a machine with code written in C and D to perform the exact same operations. The code was then compiled using different compilers and using different arguments on each compiler to figure out which was the best way to go about using D versus C++. The results do shed light on D's claims on performance. The results are mentioned below and you can visit <http://dgit.in/DversusC> for the complete discussion on the same.

D (const ref arguments, dmd compiler, correct optimization): 9.6s

D (in arguments, dmd compiler, correct optimization): 6.1s

D (in arguments, gdm compiler, correct optimization): 3.9s

C++: 3.8s

We see that the default compiler DMD isn't that great, especially when it adds 2.2 seconds to the overall runtime despite having the correct optimization and the right arguments being passed to the compiler. However, when an alternative compiler (gdm) was used with the proper arguments and all optimization done, D performed just as good as C++ does. So would you say it's worth the effort?

Resources and further reading:

- <http://dlang.org/download.html>
- <http://dlang.org/spec.html>
- http://www.tutorialspoint.com/d_programming/index.htm
- <http://www.dprogramming.com/tutorial.php>
- <http://ddili.org/ders/d.en/index.html>
- <http://www.prowiki.org/wiki4d/wiki.cgi?FrontPage>
- <http://www.wikiservice.at/d/wiki.cgi?EditorSupport>
- <http://dui.sourceforge.net/>
- <http://www.dsource.org/projects/build/>
- <http://www.dprogramming.com/dfl.php> >

*the source code



Learn how Bootstrap 3 works

>>Bootstrap Grid 3 is one of the most commonly used libraries for making fluid layouts for websites. Here is a good article explaining the working behind the framework.

<http://dgit.in/Bootstrap3Grid>



Shorten map locations using Map Code

>>Mapcode is a way to shorten any location down to just 4 letters and a dot. For more complex addressed they've 5 letters and a dot. Here's how you can use it in your project.

<http://dgit.in/MapCode>



Smoother fonts with DirectWrite

>>The next version of Google Chrome will enable DirectWrite instead of Graphics Device Interface for crisp rendering of text even on high DPI displays.

<http://dgit.in/ChromeDirectWrite>